

Closing the near-gap residuals: convention exactness with explicit margins and the split-alignment rearrangement identity

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

Unit U10 protected the near-gap regime by the operator-monotonicity lemma, leaving two named one-page residuals. This note closes both. Scope: the chain's stated convention and matched bookkeeping, all certified intensities.

2. R-U10-1: convention exactness with explicit margins

The convention $\hat{r} = r_R + 2\lambda'I$ uses the SELF-CONSISTENT effective slope: $\lambda' = 3u + 30vM = -2.58 + 97.2 \times 0.109414 = 8.0551$, and the artefact's calibration block verifies it directly at the LAM point ($A = 0.01$, $I_{\text{cal}} = 10^{-4}$): measured dressed mass $0.3061321 = r_R + 2\lambda'I_{\text{cal}}$ with second-order discrepancy

$$\delta_{2\text{nd}} = 6.7 \times 10^{-8} \quad \left(\text{relative } \frac{6.7 \times 10^{-8}}{1.61 \times 10^{-3}} = 4.2 \times 10^{-5} \right).$$

The discrepancy is the quadratic Taylor remainder of the self-consistent map $I \mapsto \hat{r}(I)$, hence scales as $(I/I_{\text{cal}})^2$:

$$\delta_{2\text{nd}}(I = 2 \times 10^{-3}) \leq 6.7 \times 10^{-8} \times 20^2 = 2.7 \times 10^{-5} \ll \underbrace{\sim 2\lambda'M_R I = 3.5 \times 10^{-3}}_{\text{lemma's linear protection at } 2 \times 10^{-3}},$$

a $\times 130$ margin floor at the worst certified intensity (and $\times 1300$ at the production anchor, where the remainder is 1.1×10^{-6} vs 7.0×10^{-4}). Class representativity: the self-consistent map's curvature is built from u, v , and the isotropic response functions $M, M' = -J(0), M''$ of the dressed kernel — objects that depend on the pattern only through the total intensity I (the pattern enters $\hat{r}$ through I alone in the convention, and the response functions are isotropic integrals); the LAM-point calibration therefore bounds the remainder for EVERY pattern at the same I up to the stated scaling. CONCLUSION: the convention's slack cannot disturb the U10 positivity at any certified intensity — the inequality $H_{\mathcal{P}} \geq D_0$ holds with $O(I^2)$ slack bounded $\times 130$ below the linear gap it would need to close.

3. R-U10-2: the split-alignment rearrangement identity

Exact identities, no estimates: (i) $D + W = D_0 + 2\lambda'I \text{Id} + W_{\text{offd}}$ (the convention plus the P^2 representation); (ii) $\text{tr } X = 0$ for $X = D^{-1/2}W_{\text{offd}}D^{-1/2}$ (strictly off-diagonal in the plane-wave

basis), so the chain’s Lemma-B object is $\delta F_{\text{off}} = \frac{1}{2V} \text{tr} \ln(\mathbf{1} + X)$ exactly. Therefore

$$\underbrace{\frac{1}{2V} \text{tr} [\ln(D_0 + \lambda' P^2) - \ln D_0]}_{\text{U10 total trace}} = \underbrace{\frac{1}{2V} \text{tr} [\ln D - \ln D_0]}_{\text{diagonal Hartree gain}} + \underbrace{\frac{1}{2V} \text{tr} \ln(\mathbf{1} + X)}_{\delta F_{\text{off}}},$$

an algebraic rearrangement (insert and remove $\ln D$). The chain’s master decomposition $\Delta F = \Phi + \Delta F_0 + \delta F_{\text{off}}$ books the diagonal gain inside the layer excursion ΔF_0 (it is the fluctuation part of $F_0(\hat{r}) - F_0(r_R)$; the algebraic remainder of ΔF_0 is the variational envelope structure that Prop A’s claim block handles); the lemma’s decomposition $\Delta F = \Phi + (\text{algebraic remainder}) + (\text{total trace})$ groups the same three trace pieces differently. Both are partitions of the SAME $F[\mathcal{P}] - F[\mathcal{R}_H]$: no term is counted twice (each $\text{tr} \ln$ difference appears in exactly one group of each partition) and none is dropped (the identity above accounts for all three). The selection argument may therefore use the chain’s split on the band intervals (where the P_B floors are needed) and the lemma’s split on interval I' (where the total trace is nonnegative structurally), with the regime boundary at $M = M_+$ inherited from Prop A’s own interval decomposition. Regularization footnote: all trace differences above are of the form $\text{tr}[\ln(A + B) - \ln A]$ with B bounded ($\|\lambda' F^2\|_\infty \leq 4\lambda' S^2$ finite) and A^{-1} of k^{-4} decay — the standard finite log-determinant ratio per unit volume that the chain’s Lemma B already manipulates.

4. Consequence

G-U1-SMALLT reduces to the SINGLE registered machine check R-U10-3 (numerically verify the trace inequality and its small- I expansion on test patterns). Upon its execution the U9 flag lifts entirely and the U1 T6-entry proposal returns to “operator sign-off required” with no open mathematical condition.

5. Numerical verification

The only numbers used are certified artefact values ($\delta_{2\text{nd}} = 6.7 \times 10^{-8}$, the calibration shift 1.61×10^{-3} , $\lambda' = 8.0551$, $M_R = 0.109414$) plus exact scaling arithmetic (20^2 ; $2\lambda' M_R = 1.763$). R-U10-3 remains the registered machine check (not executable this session; sandbox shell unavailable).

Quantitative sanity check (mandatory): magnitude — the protection- to-slack ratios ($\times 1300$ anchor, $\times 130$ endpoint) shrink exactly as $1/I$ (linear protection vs quadratic slack), so the endpoint is the binding case and is quoted as the floor; dimensional — $\delta_{2\text{nd}}$ and $2\lambda' M_R I$ are both mass²-like dressing shifts (the comparison is homogeneous); limit case — as $I \rightarrow I_{\text{cal}}$ the scaled remainder reproduces the measured 6.7×10^{-8} identically (scaling factor 1); consistency — the rearrangement identity of section 3 was verified term-by-term against the chain’s Lemma-B display ($\text{tr} X = 0$ is the off-diagonality of W_{offd} , an exact structural fact, not a numerical one).

6. Devil’s-advocate (three objections; CLAUDE.md 6.3)

- (α) “The $(I/I_{\text{cal}})^2$ scaling of the remainder assumes the self-consistent map is analytic with the quadratic term dominant through $I = 2 \times 10^{-3}$; a cubic term could spoil it.” VALID-with-mitigation: the map’s expansion parameter is $2\lambda' I/\hat{r} \leq 0.096$ at the endpoint, so the cubic term is suppressed by another factor ≤ 0.1 relative to the quadratic — folding it in changes the $\times 130$ floor by at most 10%. The R-U10-3 script can verify the remainder directly at the endpoint (one extra self-consistent solve) at negligible cost; added to its registered spec.

- (β) “Class representativity leans on ‘the pattern enters only through I ’ — but λ' itself contains $M(\hat{r})$ and different patterns at the same I have identical $\hat{r}$ only BY the convention; the argument is circular.” DISMISSED with the convention’s content: within the matched bookkeeping the isotropic dressing IS defined by the $(I, \hat{r})$ self-consistency — that is hypothesis H-LAYER’s layer framing, already a named hypothesis of the U1 set; GIVEN the layer, the dressing of two patterns with equal I coincides by construction, and the remainder bound is class-uniform. The objection re-discovers H-LAYER, which is tracked, not hidden.
- (γ) “Section 3 asserts the diagonal gain is booked inside ΔF_0 without exhibiting the chain’s F_0 definition line-by-line.” VALID-with-mitigation: the assertion is the natural reading (the layer’s $F_0(m)$ is the isotropic-Gaussian free energy whose fluctuation part is $\frac{1}{2V} \text{tr} \ln(m + C(\cdot)^2)$, Math437’s slaved family), and the rearrangement identity is independent of WHERE the chain books the diagonal gain (any booking is one of the partitions). The line-by-line exhibit against the Math424-AddA production decomposition folds into the R-U10-3 script’s companion comment block — registered, and the identity’s validity does not wait on it.

7. Result footer

Result ID:	B1-RH-ENUM / neargap-residual-closure (unit U11; closes R-U10-1 + R-U10-2)
Precise statement:	(i) R-U10-1 CLOSED at T4 grade: the convention’s $O(I^2)$ slack is bounded by the calibrated quadratic remainder scaling -- $2.7e-5$ at the endpoint, x130 below the U10 linear protection (x1300 at the anchor); class-representative via the isotropic response structure. (ii) R-U10-2 CLOSED by exact algebra: the chain’s split and the lemma’s split are rearrangements of the same $F[P] - F[R_H]$ ($\text{tr } X = 0$; insert/remove $\ln D$); no double-count, no dropped term. G-U1-SMALLT reduces to the single machine check R-U10-3.
Scope:	Chain convention + matched bookkeeping; all certified intensities; amended class.
Dependencies:	U10 lemma; artefact lam_calibration block ($\text{delta_2nd} = 6.7e-8$); P^2 representation; Prop A interval structure.
Evidence grade:	ANALYTIC (exact algebra) + CALIBRATED (certified artefact values + scaling).
Reproduction command:	(registered R-U10-3, spec extended) trace-inequality check + endpoint self-consistent remainder solve.
Expected output:	trace difference ≥ 0 ; remainder at $2e-3$ within 10% of $2.7e-5$; JSON under claims/B1-RH-ENUM/runs/.
Falsification gate:	Measured endpoint remainder exceeding $\sim 3e-5$ by more than the cubic allowance

(would reopen R-U10-1); any term of the rearrangement identity failing the line-by-line exhibit (would reopen R-U10-2).

Tier before / after: No tier action. B1 stays T5. The U1 proposal condition is now: R-U10-3 machine check + operator sign-off (no open mathematical condition).

No-overclaim statement: R-U10-3 is still pending (no machine execution this session); the U9 flag lifts only upon it. The closures here are at the stated grades (calibrated scaling; exact algebra), not unconditional theorems. PDF build + linter PENDING operator-side.

Next required action: Execute R-U10-3 (one small script: trace inequality on test patterns + endpoint remainder + line-by-line booking exhibit); then operator decision on U1. Mainline continues (U12).